

Sobolev maps to manifolds: the distributional Jacobian and its applications

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Outline of this presentation

- 1 Introduction
- 2 What is the role of the Jacobian?
 - The distributional Jacobian
 - When $\mathcal{N} = \mathbb{S}^k$
 - From sphere to more general manifold: how?
- 3 The distributional Jacobian for low regularity maps

Definition

Consider the space

$$W^{s,p}(\mathbb{B}^N; \mathcal{N}) := \{f \in W^{s,p}(\mathbb{B}^N) : f(x) \in \mathcal{N}\},$$

where $\mathcal{N}$ is an embedded closed manifold.

Example (Schoen and Uhlenbeck, (83'))

Let $2 \leq p < 3$. The map $f \in W^{1,p}(\mathbb{B}^3; \mathbb{S}^2)$ defined by

$$f(x) := \frac{x}{|x|}, \forall x \in \mathbb{B}^3 \setminus \{0\}$$

cannot be approximated with smooth maps in $C^\infty(\overline{\mathbb{B}^3}; \mathbb{S}^2)$.

Key element: $\pi_2(\mathbb{S}^2) \neq \{0\}$.

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When is $C^\infty(\overline{\mathbb{B}^N}; \mathcal{N})$ dense in $W^{s,p}(\mathbb{B}^N; \mathcal{N})$?

Assume that $0 < s < \infty$ and $1 \leq p < \infty$.

Theorem

$C^\infty(\overline{\mathbb{B}^N}; \mathcal{N})$ is dense in $W^{s,p}(\mathbb{B}^N; \mathcal{N})$ if and only if either (i) $sp \geq N$; or (ii) the $\lfloor sp \rfloor$ -th homotopy group $\pi_{\lfloor sp \rfloor}(\mathcal{N}) \simeq \{0\}$.

- $s = 1$: Bethuel (91') (using an argument of *good* cubes and *bad* cubes);
- $s \geq 1$ integer: Bousquet, Ponce and Van Schaftingen (15');
- $0 < s < 1$: Brezis and Mironescu (15');
- $s \geq 1$ non-integer: Demailly (25');
- General domain: Hang and Lin (03').

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When strong density fails

Problem

If the strong density fails, how to characterize the $W^{s,p}$ maps which can be strongly approximated with smooth maps?

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Jacobian structure

Recall that for $f \in C^1(\mathbb{R}^N; \mathbb{R}^N)$,

$$\det(\nabla f) = \sum_{j=1}^N (\partial_j f_i) C_{i,j}, \quad i \in \{1, \dots, N\},$$

where $C_{i,j}$ is the co-factor of $\partial_j f_i$ in the matrix $(\partial_\ell f_k)_{\ell,k}$.

Moreover, if $f \in C^2(\mathbb{R}^N; \mathbb{R}^N)$, then

$$\det(\nabla f) = \sum_{j=1}^N \partial_j (f_i C_{i,j}), \quad i \in \{1, \dots, N\},$$

which implies that the Jacobian has a **divergence structure**.

Another way to see this structure is in terms of **differential forms**:

$$\begin{aligned}\det(\nabla f) \, dx_1 \wedge \cdots \wedge dx_N &= df_1 \wedge \cdots \wedge df_N \\ &= d[(-1)^{i-1} f_i df_1 \wedge \cdots \wedge \widehat{df_i} \wedge \cdots \wedge df_N].\end{aligned}$$

Definition (Ball (76'))

Given $f \in (W^{1,N-1} \cap \mathcal{L}^\infty)(\Omega; \mathbb{R}^N)$, we define its **distributional Jacobian** $\text{Det}(\nabla f)$ in the sense of distributions by

$$\begin{aligned}\langle \text{Det}(\nabla f), \xi \rangle &:= - \sum_{j=1}^N \int_{\Omega} f_1 C_{1,j} \partial_j \xi = - \frac{1}{N} \sum_{j=1}^N \int_{\Omega} \left(\sum_{i=1}^N f_i C_{i,j} \right) \partial_j \xi \\ &= \frac{(-1)^{N+1}}{N} \int_{\Omega} \left(\sum_{i=1}^N (-1)^{i-1} f_i \widehat{df_i} \right) \wedge d\xi, \quad \forall \xi \in C_c^\infty(\Omega).\end{aligned}$$

The case where $\mathcal{N} = \mathbb{S}^k$

Let $k \leq p < k + 1$. For each

$$f \in W^{1,p}(\Omega^{k+1}; \mathbb{S}^k),$$

we have

$$\begin{aligned} \langle \text{Det}(\nabla f), \xi \rangle &= \frac{(-1)^k}{k+1} \int_{\Omega} \left(\sum_{i=1}^{k+1} (-1)^{i-1} f_i \widehat{df}_i \right) \wedge d\xi \\ &= \frac{(-1)^k}{k+1} \int_{\Omega} f^* \omega_{\mathbb{S}^k} \wedge d\xi, \end{aligned}$$

where $\omega_{\mathbb{S}^k}$ is the volume form of $\mathbb{S}^k$.

Det(∇f) "detects" singularity

Example

Let $2 \leq p < 3$. Let $f \in W^{1,p}(\mathbb{B}^3; \mathbb{S}^2)$ be defined as

$$f(x) := \frac{x}{|x|}, \forall x \in \mathbb{B}^3 \setminus \{0\}.$$

For each $\xi \in C_c^\infty(\Omega)$, we have

$$\begin{aligned} \langle \text{Det}(\nabla f), \xi \rangle &= \frac{1}{3} \int_{\mathbb{B}^3} f^* \omega_{\mathbb{S}^2} \wedge d\xi = \lim_{r \rightarrow 0} \frac{1}{3} \int_{\mathbb{B}^3 \setminus B_r} f^* \omega_{\mathbb{S}^2} \wedge d\xi \\ &= -\frac{1}{3} \xi(0) \lim_{r \rightarrow 0} \int_{\mathbb{S}_r} f^* \omega_{\mathbb{S}^2} = -\frac{4\pi}{3} \xi(0), \end{aligned}$$

thus $\text{Det}(\nabla f) = -\frac{4\pi}{3} \delta_0$.

Theorem (Bethuel (90'), Bethuel, Coron, Demengel, Hélein (91'))

Let $k \leq p < k + 1$. A map $f \in W^{1,p}(\mathbb{B}^{k+1}; \mathbb{S}^k)$ can be approximated with smooth maps if and only if $\text{Det}(\nabla f) = 0$.

Sketch of the proof.

" $\implies$ " $\langle \text{Det}(\nabla f), \xi \rangle = \lim_{\ell \rightarrow \infty} \langle \text{Det}(\nabla f_\ell), \xi \rangle = 0$, provided $C^\infty \ni f_\ell \rightarrow f$.

" $\impliedby$ " Using suitable test functions, we obtain

$$\int_{\partial Q_r(x)} f^* \omega_{\mathbb{S}^k} = 0, \text{ for a.e. } x \in \mathbb{B}^{k+1}, \text{ a.e. } r > 0.$$

Hence, by degree theory we have

$$f|_{\partial Q_r(x)} \text{ is nullhomotopic,}$$

and this implies that f can be approximated with smooth maps. □

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To lower dimensional spheres

Definition (Brezis, Coron, Lieb (86'))

Let $k \leq p < k + 1$ and $N \geq k + 1$. Given $f \in W^{1,p}(\Omega; \mathbb{S}^k)$, we define its **distributional Jacobian** $\text{Det}(\nabla f)$ as an $(N - k - 1)$ -current by

$$\langle \text{Det}(\nabla f), \xi \rangle := \frac{(-1)^k}{k+1} \int_{\Omega} f^* \omega_{\mathbb{S}^k} \wedge d\xi, \forall \xi \in C_c^\infty(\Omega; \Lambda^{N-k-1}).$$

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Consider a more general manifold $\mathcal{N} \subset \mathbb{R}^{\nu}$

- Replacing $\omega_{\mathbb{S}^k}$ by all closed k -forms ω on $\mathcal{N}$.
- Assume $\mathcal{N}$ is such that, for each $f \in C^\infty(\mathbb{S}^k; \mathcal{N})$,

$$\left[\int_{\mathbb{S}^k} f^* \omega = 0, \forall \text{ closed } k\text{-form } \omega \text{ on } \mathcal{N} \right] \implies f \text{ is nullhomotopic.}$$

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Further contributions

- Hardt and Rivière (03', 08'): using scans;
- Bousquet, Ponce and Van Schaftingen (25'): $s \in \mathbb{N}_*$;
- Canevari and Orlandi (26'): using flat chains with values into a group.

The low regularity case $0 < s < 1$.

Now we turn to the case of $W^{s,p}(\mathbb{B}^N; \mathcal{N})$ maps with $k \leq sp < k+1$. Can we extend the above Theorem?

- What is “ $\int_Q f^* \omega \wedge d\xi = 0$ ”, for a map in $W^{s,p}(\mathbb{B}^N; \mathcal{N})$?
- What is “ $\int_{\partial Q^k} f^* \omega$ ” for $f \in W^{s,p}(\partial Q^k; \mathcal{N})$?
- What is the meaning of “nullhomotopy” for a map in $W^{s,p}(\partial Q^k; \mathcal{N})$?

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Definition

It is natural to consider the operator (at least formally) given by

$$\langle Tf, \zeta \rangle := \int_{\mathbb{R}^N} f^* \omega \wedge d\zeta, \forall \zeta \in \text{Lip}(\mathbb{R}^N; \Lambda^{N-k-1}).$$

If $\zeta \in C_c^\infty(\mathbb{R}^N; \Lambda^{N-k-1})$, then we formally have

$$\int_{\mathbb{R}^N} f^* \omega \wedge d\zeta = - \int_{\mathbb{R}^N \times (0, \infty)} d[\widetilde{F}^* \alpha \wedge d\widetilde{\zeta}] = - \int_{\mathbb{R}^N \times (0, \infty)} \widetilde{F}^*(d\alpha) \wedge d\widetilde{\zeta}.$$

Here, $\widetilde{F} := \widetilde{\Pi} \circ F$, with F an extension of f , and $\widetilde{\zeta}(x, t) := \zeta(x)$, $\forall x \in \mathbb{R}^N$, $\forall t \geq 0$.

Theorem (Daille, Mironescu, X. (25'))

Let $k \geq 2$. Assume that $0 < s < 1$ and $sp = k < N$.

- (1) The map T , defined for $f \in W_1^{1,k}(\mathbb{R}^N; \mathcal{N})$, has an (unique) extension by continuity to $W_1^{s,p}(\mathbb{R}^N; \mathcal{N})$.
- (2) The extension, still denoted T , satisfies

$$|\langle Tf, \zeta \rangle| \leq C |f|_{W^{s,p}}^p |\zeta|_{\text{Lip}}, \quad \forall f \in W_1^{s,p}(\mathbb{R}^N; \mathcal{N}), \forall \zeta \in \text{Lip}(\mathbb{R}^N; \Lambda^{N-k-1}).$$

- (3) Let $\tilde{\Pi}$ be an almost projection and $\alpha \in C_c^\infty(\mathbb{R}^n; \Lambda^k)$ be an extension of ω . We have

$$\langle Tf, \zeta \rangle = - \int_{\mathbb{R}^N \times (0, \infty)} \tilde{F}^*(d\alpha) \wedge d\tilde{\zeta}, \quad \forall \zeta \in \text{Lip}(\mathbb{R}^N; \Lambda^{N-k-1}).$$

Here, $\tilde{F} := \tilde{\Pi} \circ F$, with F an extension of f , and $\tilde{\zeta}(x, t) := \zeta(x)$, $\forall x \in \mathbb{R}^N$, $\forall t \geq 0$.

T “hears” singularities

Consider the class

$$\begin{aligned} \mathcal{R}_1 := \{f: \mathbb{R}^N \rightarrow \mathcal{N}: f \text{ is constant in } \mathbb{R}^N \setminus \mathbb{B}^N, f \in C^\infty(\overline{\mathbb{B}^N} \setminus \mathcal{S}(f)), \\ \mathcal{S}(f) \text{ is a } (N - k - 1)\text{-closed, oriented submanifold} \\ \text{of } \mathbb{B}^N, |\nabla f(x)| \leq C(f)/\text{dist}(x, \mathcal{S}(f)), \forall x \in \mathbb{R}^N\}. \end{aligned}$$

Theorem (Detaile, Mironescu, X. (25'))

Let $f \in \mathcal{R}_1$. Then

$$\langle Tf, \zeta \rangle = (-1)^{k(N+1)+1} \sum_{i=1}^{\ell} c_i \int_{\mathcal{S}_i} \zeta, \forall \zeta \in \text{Lip}(\mathbb{R}^N; \Lambda^{N-k-1}),$$

where $c_i := \int_{S_\varepsilon(z)} f^* \omega, \forall z \in \mathcal{S}_i, \forall 0 < \varepsilon < \bar{\varepsilon} = \bar{\varepsilon}(\mathcal{S})$.

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Slicing: From $\mathbb{R}^N$ to $\mathbb{R}^{k+1}$

Lemma

Let $\rho = \rho(y)$ be a standard mollifier and set $\rho_{\varepsilon, y_0}(y) := \rho_{\varepsilon}(y - y_0)$, $\forall y, y_0 \in \mathbb{R}^v$. Then, for a.e. $y_0 \in \mathbb{R}^v$, we have

$$\langle Tf(\cdot, y_0), \xi \rangle = \sigma \lim_{\varepsilon \rightarrow 0} \langle Tf, \xi \wedge (\rho_{\varepsilon, y_0} dy) \rangle, \forall \xi = \xi(x) \in \text{Lip}(\mathbb{R}^{\ell}; \Lambda^{\ell-k-1}).$$

Slicing: From $\mathbb{R}^{k+1}$ to ∂Q^{k+1}

Lemma

Let $f \in W^{s,p}(\mathbb{R}^{k+1}; \mathcal{N})$. For $\varepsilon > 0$ and $0 < \eta \leq \varepsilon/2$, let $\psi_\eta = \psi_{\varepsilon,\eta}$ be defined by

$$\psi_\eta(r) = \psi_{\eta,\varepsilon}(r) := \begin{cases} 1, & \text{if } r \leq \varepsilon - \eta \\ (\varepsilon - r)/\eta, & \text{if } \varepsilon - \eta \leq r \leq \varepsilon \\ 0, & \text{if } r \geq \varepsilon \end{cases}$$

Let $\eta_\ell \rightarrow 0$. Then, for a.e. $\varepsilon > 0$, we have

$$\int_{P+\partial Q_\varepsilon} f^* \omega = \lim_{\ell \rightarrow \infty} (-1)^{k+1} \langle Tf, \psi_{\eta_\ell}(|\cdot - P|_\infty) \rangle, \text{ for a.e. } P \in \mathbb{R}^{k+1}.$$

A characterization of the closure of smooth maps

Theorem (Detaile, Mironescu, X. (25'))

Assume that (i) $k \leq sp < k + 1 \leq N$; (ii) the closed manifold $\mathcal{N}$ satisfies

$$\left[\int_{\mathbf{S}^k} g^* \omega = 0, \forall \text{ smooth closed } k\text{-form } \omega \text{ on } \mathcal{N} \right] \implies g \text{ is nullhomotopic.}$$

Let $f \in W_1^{s,p}(\mathbb{R}^N; \mathcal{N})$. Then

$$f \in \overline{C_1^\infty(\mathbb{R}^N; \mathcal{N})}^{W_1^{s,p}} \Leftrightarrow \forall \text{ smooth closed } k\text{-form } \omega \text{ on } \mathcal{N}, T_\omega f = 0.$$

Thank you for your attention!